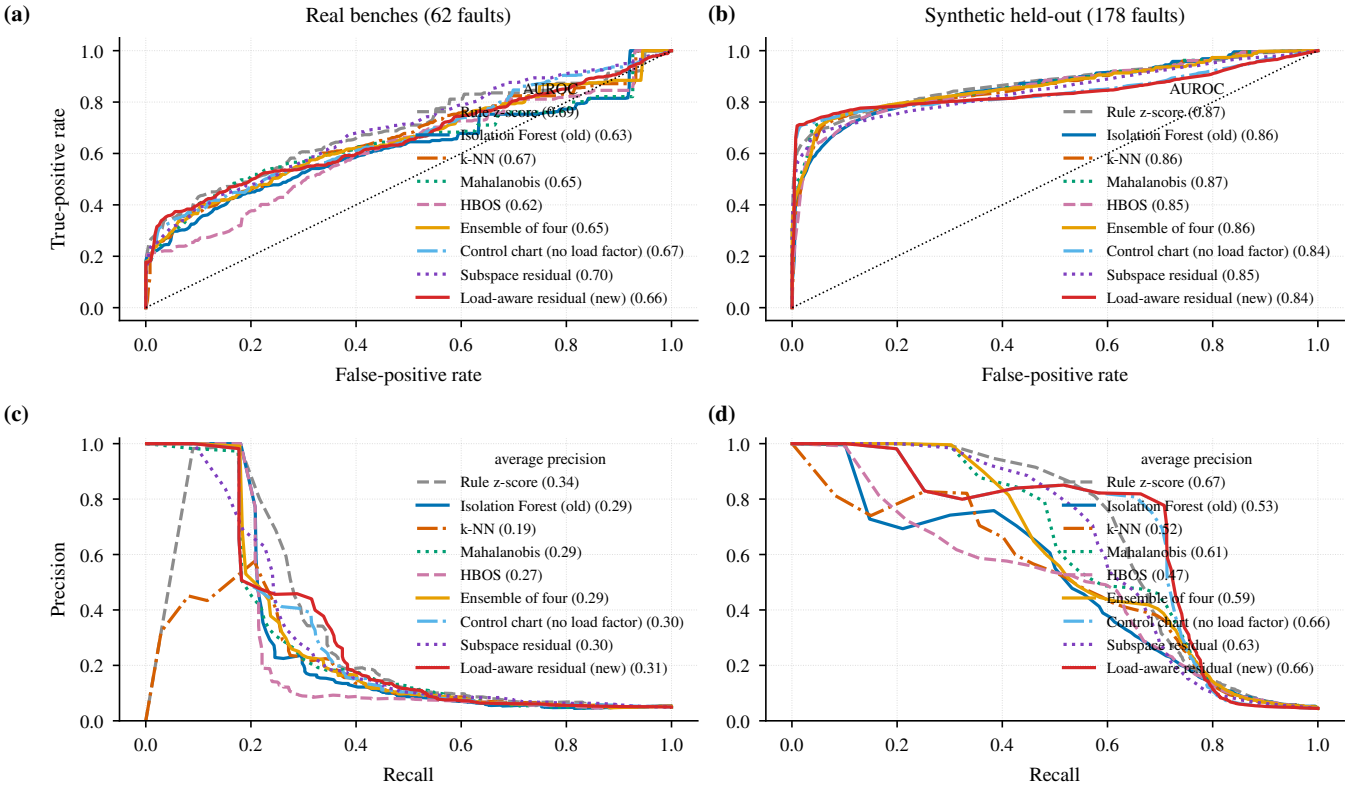
(a)
Real benches (62 faults)
True-positive rate
False-positive rate
AUROC
Rule z-score (0.69)
Isolation Forest (old) (0.63)
k-NN (0.67)
Mahalanobis (0.65)
HBOS (0.62)
Ensemble of four (0.65)
Control chart (no load factor) (0.67)
Subspace residual (0.70)
Load-aware residual (new) (0.66)
(b)
Synthetic held-out (178 faults)
False-positive rate
AUROC
Rule z-score (0.87)
Isolation Forest (old) (0.86)
k-NN (0.86)
Mahalanobis (0.87)
HBOS (0.85)
Ensemble of four (0.86)
Control chart (no load factor) (0.84)
Subspace residual (0.85)
Load-aware residual (new) (0.84)
(c)
Precision
Recall
average precision
Rule z-score (0.34)
Isolation Forest (old) (0.29)
k-NN (0.19)
Mahalanobis (0.29)
HBOS (0.27)
Ensemble of four (0.29)
Control chart (no load factor) (0.30)
Subspace residual (0.30)
Load-aware residual (new) (0.31)
(d)
Recall
average precision
Rule z-score (0.67)
Isolation Forest (old) (0.53)
k-NN (0.52)
Mahalanobis (0.61)
HBOS (0.47)
Ensemble of four (0.59)
Control chart (no load factor) (0.66)
Subspace residual (0.63)
Load-aware residual (new) (0.66)